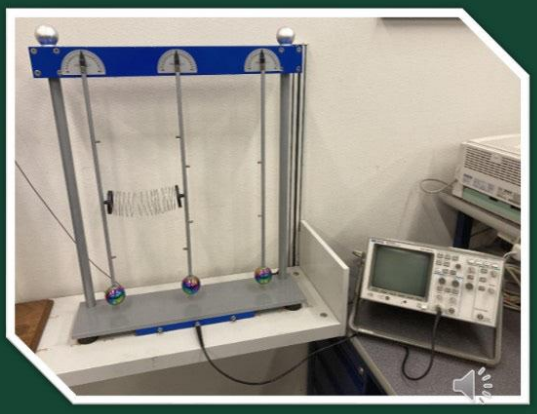
CAL POLY

LAB 5: DOUBLE PENDULUM

GABRIEL KEITH CHOI – ERIC TRAN

MOHAMMAD HASAN



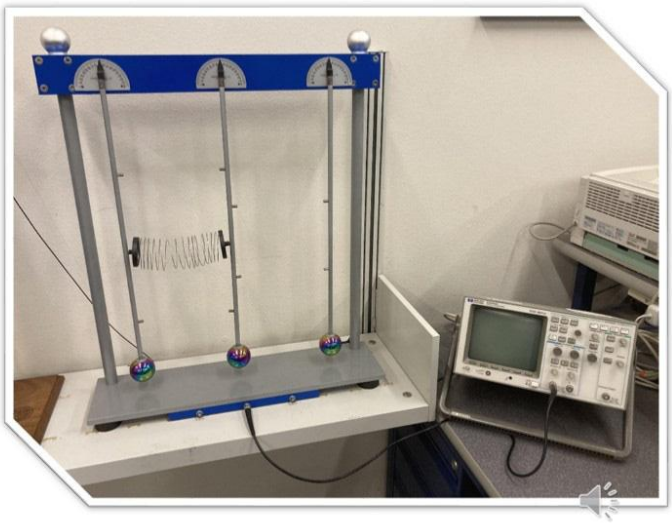
EXPERIMENT OBJECTIVES

- PERFORM FREE VIBRATION MODAL TESTING OF THE DOUBLE PENDULUM SYSTEM IN ITS TWO SINGLE DEGREE OF FREEDOM (SDOF) CONFIGURATIONS
- DETERMINE THE FIRST TWO MODAL (NATURAL) FREQUENCIES FOR THE TWO DEGREES OF FREEDOM PENDULUM SYSTEM
- STUDY THE EFFECT OF CONSTRUCTIVE AND DESTRUCTIVE INTERFERENCE RESULTING IN THE VIBRATORY PHENOMENON KNOWN AS BEATS.
- DEVELOP A COMPUTER SIMULATION MODEL OF THE TWO DEGREE OF FREEDOM DOUBLE PENDULUM SYSTEM UNDERGOING THE BEAT PHENOMENON
- COMPARE THE RESULTS OF THE SIMULATION MODEL TO THE EXPERIMENTAL BEAT PHENOMENON DATA
- UNDERSTAND AND EXPLAIN THE CAUSE OF BEATS DUE TO ASYNCHRONOUS VIBRATION

EXPERIMENTAL HARDWARE

EQUIPMENT USED:

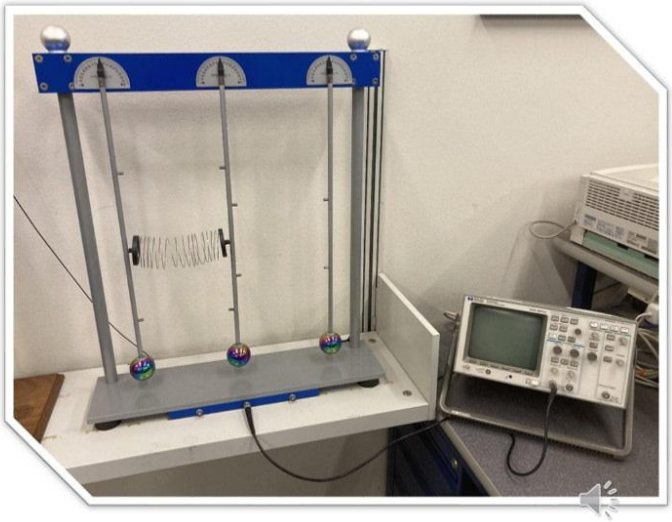
- TRIPLE PENDULUM APPARATUS
  - DETACHABLE SPRINGS
  - MAGNETIC ROTARY POTENTIOMETERS
  - BNC CONNECTOR PORT
- OSCILLOSCOPE
  - KEYSIGHT DSOX1204G
    - THE OSCILLOSCOPE SHOWN IN THE PHOTO IS NOT THE ONE USED FOR OUR EXPERIMENT



SCHEMATIC OF EXPERIMENTAL SETUP

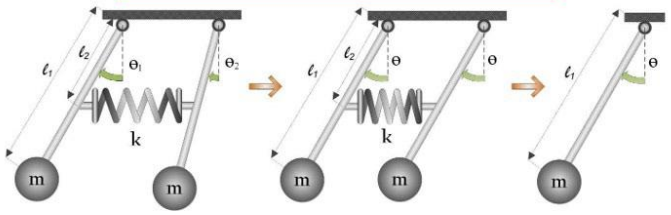
CONNECTIONS:

- THE SPRING IS ATTACHED TO THE LEFT AND CENTER PENDULUMS AT PEGS HALFWAY DOWN
- A CABLE CONNECTS THE APPARATUS' BNC PORT AT THE BOTTOM TO THE INPUT PORT OF THE OSCILLOSCOPE
- THE OSCILLOSCOPE IS CONNECTED TO A POWER SOURCE

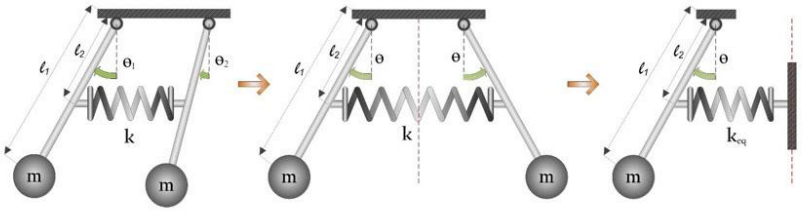


MODAL ANALYSIS TESTING (MODE 1 AND MODE 2)

MODE 1: IDENTICAL DISPLACEMENT



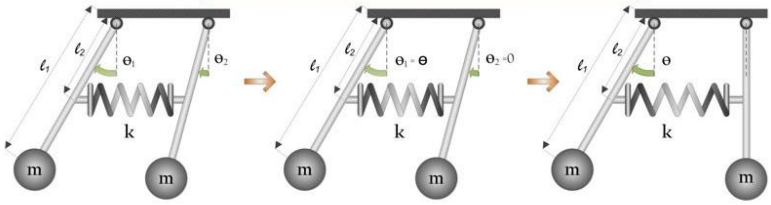
MODE 2: MIRRORED DISPLACEMENT



CAL POLY

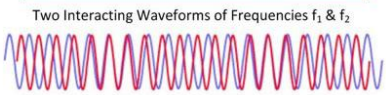
MODAL ANALYSIS TESTING (BEATS)

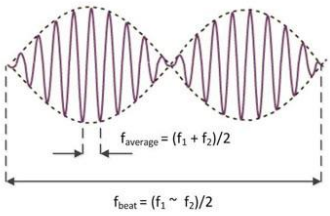
BEAT PHENOMENON DISPLACEMENT



BEAT BEHAVIOR VISUALIZED

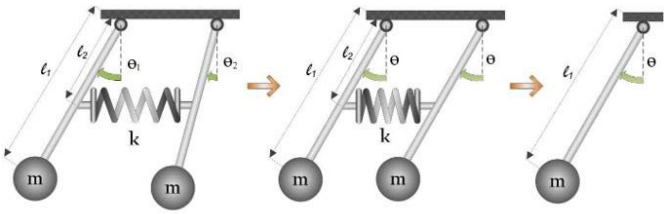
Two Interacting Waveforms of Frequencies f_1 & f_2

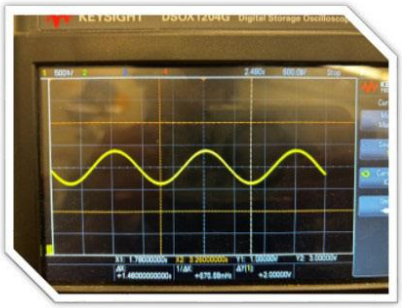




CAL POLY

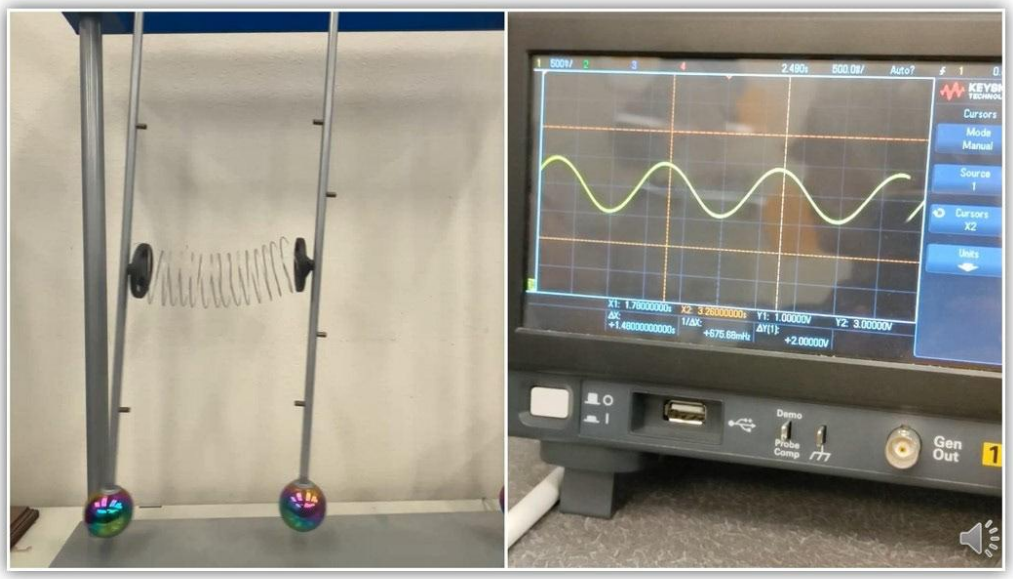
MODAL ANALYSIS TESTING 1 (FIRST NATURAL FREQUENCY)



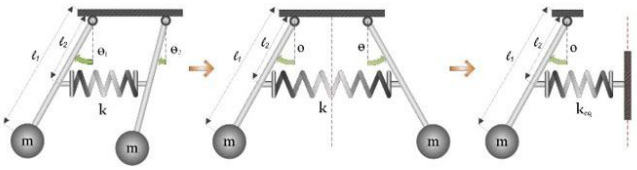


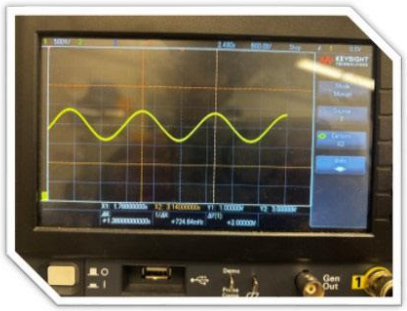
	Theoretical Value (mHz)	Experimental Value (mHz)
First Natural Modal Frequency (ω_{n1})	688 mHz	675.68 mHz

MODAL ANALYSIS TESTING 1 (FIRST NATURAL FREQUENCY)



MODAL ANALYSIS TESTING 2 (SECOND NATURAL FREQUENCY)

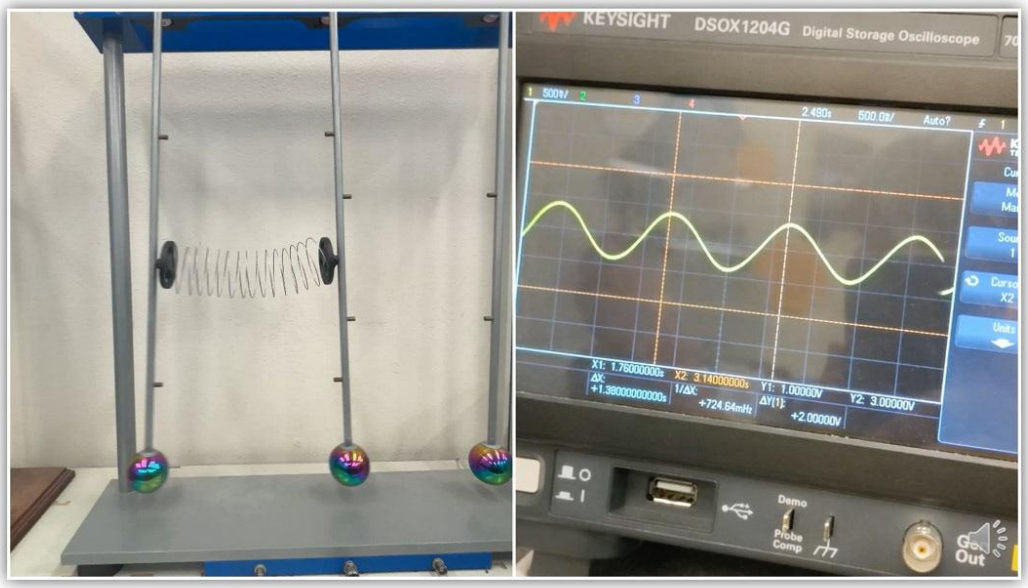




	Experimental Value (mHz)
Second Natural Modal Frequency (ωn2)	724.64 mHz

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MODAL ANALYSIS TESTING 2 (SECOND NATURAL FREQUENCY)



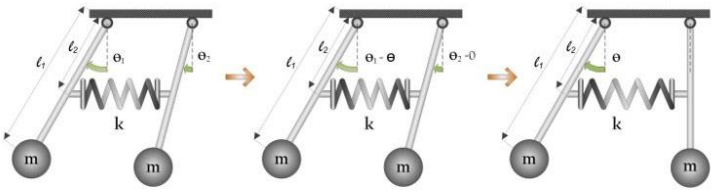
CAL POLY

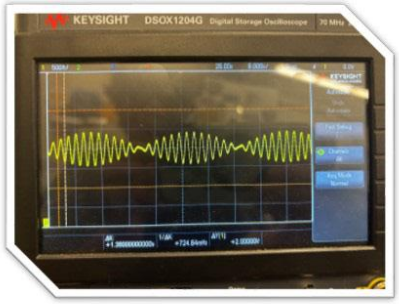
SYSTEM IDENTIFICATION (SPRING CONSTANT)

Quantity	Expression
Natural Frequency	$sqrt\sqrt{\frac{2kL_2^2 - mgL_1}{mL_1^2}}$
Spring Constant (Eqn.)	$\frac{\omega_n^2(mL_1^2) - mgL_1}{2L_2^2}$
Spring Constant (Value)	K = 1.17 N/m2

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MODAL ANALYSIS TESTING (BEAT PHENOMENON)





	Theoretical Value (mHz)	Experimental Value (mHz)
Average Frequency (ωa)	699.8 mHz	714.29 mHz
Beat Frequency (ωb)	24.82 mHz	27.5 mHz

CAL POLY

MODAL ANALYSIS TESTING (BEAT PHENOMENON)



THEORY VS. SIMULATION VS. RESULTS

	Theory	Simulation	Experimental
First mode (mHz)	688	688	675
Second mode (mHz)	N/A	722	724.64
Average (mHz)	699.8	705	714.29
Beat (mHz)	24.82	17	27.5
K	N/A	N/A	1.17

MATLAB SIMULATION

Simulation	Initial Conditions
Simulation Mode 1	[+5°,+5°,0,0]
Simulation Mode 2	[+5°,5°,0,0]
Beats	[+5°,0,0,0]



MATLAB SIMULATION

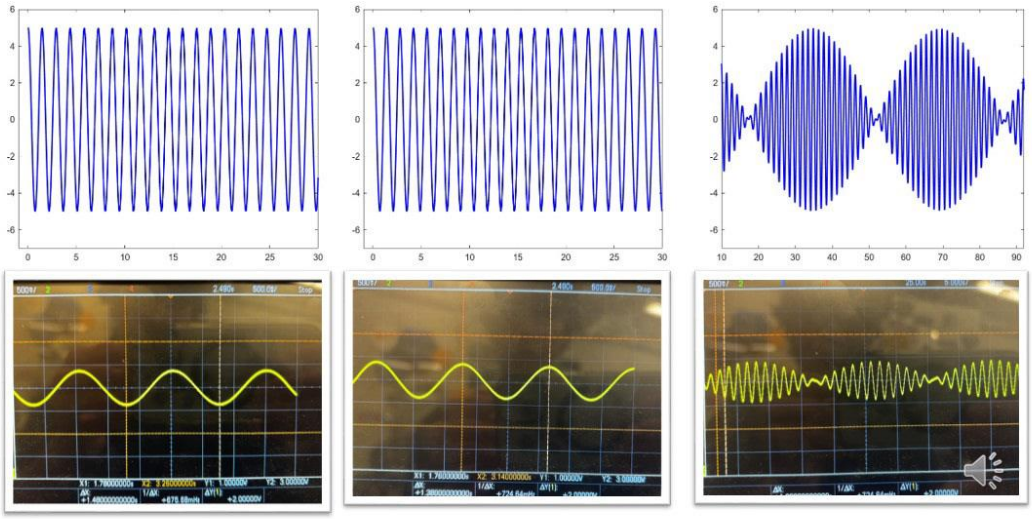
Function Script Code	Live Script Code

function dx = coupled_pendulum(t, x)
% Constants
g = 9.81; % gravity [m/s^2]
l1 = 0.525; % length of pendulum 1 [m]
l2 = 0.310; % length of pendulum 2 [m]
m = 0.51; % mass [kg]
k = 1.17; % spring constant [N/m]
 	
dx(1,1)=x(3); % dx1/dt=ẋ1
dx(2,1)=x(4); % dx2/dt=ẋ2
 	
% Second derivatives
dx(3,1) = - (g/l1)*x(1) - (k * l2^2)/(m * l1^2) * (x(1) - x(2));
dx(4,1) = - (g/l1)*x(2) - (k * l2^2)/(m * l1^2) * (x(2) - x(1));
end
	clc; clear all; close all;
	
	% Initial conditions:
	theta0 = 5 * pi / 180;
	x0 = [theta0; -theta0; 0; 0];
	tspan = [0 30];
	[t, x] = ode45(@coupled_pendulum, tspan, x0);
	
	% Plot in degrees
	Figure(1)
	plot(t, x(:,1)*180/pi, 'b-', 'LineWidth', 1.5)
	
	xlim([-1 30])
	ylim([-7 6])



SIMULATION VS. EXPERIMENT RESPONSE

Simulation 1 | Simulation 2 | Simulation 3



ERROR ANALYSIS

Case	Experimental Percent Error
Simulation 1 First Natural Modal Frequency	1.93%
Average Frequency	2.07%
Beat Frequency	10.8%

CONCLUSION AND DISCUSSION

- WE USED THE NEWTON EULER METHOD TO SYMBOLICALLY SOLVE FOR THE EQUATION OF MOTION AND THE NATURAL FREQUENCY.
- WE EXPERIMENTED WITH A PHYSICAL 2 DOF PENDULUM SYSTEM COUPLED BY A SPRING, COMPARING THE EXPERIMENTAL NATURAL FREQUENCY WITH THE ONES WE CALCULATED IN THE PRE-LAB.
- WE INVESTIGATED THE BEAT PHENOMENON, WHICH IS A BEHAVIOR THAT OCCURS WHEN ONLY ONE PENDULUM IS DISPLACED INITIALLY.
- THE BEAT IS CAUSED BY THE PENDULUMS OSCILLATING AT DIFFERENT FREQUENCIES, CAUSING PARTS OF CONSTRUCTIVE AND DESTRUCTIVE INTERFERENCE.
- WE SIMULATED THE PENDULUM USING MATLAB AND ACHIEVED MODESTLY ACCURATE VALUES CLOSE TO OUR EXPERIMENTAL DATA.
- DEVIATION FROM NORMAL MODE INITIAL CONDITIONS RESULTED IN ENERGY EXCHANGE (BEATS), AS EXPECTED FROM MODAL SUPERPOSITION.
- MULTIPLE DIFFERING DOF WILL CREATE MULTIPLE FREQUENCIES. THE BEAT PHENOMENON THEN OCCURS AND THE RESULTING OSCILLATION IS A COMBINATION OF ALL OF THE INDIVIDUAL FREQUENCIES (MODES).
- ERRORS MAY HAVE COME FROM APPLYING IDEAL ASSUMPTIONS TO REALISTIC EXPERIMENT SUCH AS SMALL ANGLE APPROXIMATION AND CONCENTRATING MASS AT THE END, WRONGING THE MOMENT OF INERTIA.
- OVERALL, WE HAD PRACTICE CONNECTING THEORETICAL, EXPERIMENTAL, AND SIMULATION DATA TOGETHER (HAND CALCULATIONS, LAB WORK, AND MATLAB RESPECTIVELY).

CHECKLIST

EXPERIMENT 5
DOUBLE PENDLUM LAB CHECKLIST

Close

Task Description	X
COMPLETE THE CHECKLIST BELOW AND INCLUDE AT THE END OF LAB REPORT	X
Setup the Hardware as per Figure 1A	X
Complete Tabular Column in Modal Analysis Test 1 Section	X
Obtain Photo of Oscilloscope Screen Showing First Natural Frequency	X
Obtain Videos V1 & V2	X
Complete All Tabular Columns in Modal Analysis Test 2 Section	X
Obtain Photo of Oscilloscope Screen Showing Second Natural Frequency	X
Obtain videos V3 & V4	X
Complete Tabular Column in Modal Analysis (Beat Phenomenon) Section	X
Obtain Two Photos of the Oscilloscope Screen Showing the Average Frequency & the Beat Frequency	X
Complete Computer Simulation Tests 1, 2, & 3 and check that the simulation responses match with those of the experiment.	X
Turn off Power to Pendulum Potentiometers (White power brick connected to power strip on the desk)	X
Turn off Oscilloscope (Push Button)	X